
Homework 1

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Chapter 1

Section 1.1

Problem 12e

Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express the proposition as an English sentence

e) $(p \rightarrow \sim r) \vee (q \rightarrow \sim r)$

If you have the flue or you miss the final examination, you will *not* pass the course.

Problem 14d,e

Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you

get an A in this class.

$$\sim q \wedge (p \wedge r)$$

e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.

$$(p \wedge q) \rightarrow r$$

Problem 32d

Construct a truth table for each of these compound propositions.

d) $(p \wedge q) \rightarrow (p \vee q)$

p	q	$(p \wedge q)$	$(p \vee q)$	$(p \wedge q) \rightarrow (p \vee q)$
F	F	F	F	T
F	T	F	T	T
T	F	F	T	T
T	T	T	T	T

Section 1.2

Problem 36

Four friends have been identified as suspects for an unauthorized access into a computer system. They have made statements to the investigating authorities. Alice said *Carlos did it*. John said *I did not do it*. Carlos said *Diana did it*. Diana said *Carlos lied when he said that I did it*.

a) If the authorities also know that exactly one of the four suspects is telling the truth, who did it? Explain your reasoning.

John was the one that did it.

Explanation:

Both Diana and Carlos cannot lie at the same time, which means one of them is telling the truth. That implies that Alice is lying, and so is John. Since John is the only one referring to himself, he is the one who did it.

b) If the authorities also know that exactly one is lying, who did it? Explain your reasoning.

Carlos did it.

Explanation:

Conversely, Carlos and Diana cannot tell the truth at the same time, and since only one suspect is lying, it means that Alice is telling the truth which implies that Carlos did it.

Section 1.3

Problem 8b

8 Use De Morgan's laws to find the negation of each of the following statements.

b) Yoshiko knows Java and calculus.

Yoshiko does not know Java, or does not know Calculus, or does not know both.

$$\sim p \vee \sim q \vee \sim r$$

Problem 10

Show that each of these conditional statements is a tautology by using truth tables.

b) $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$

p	q	r	$(p \rightarrow q)$	$(q \rightarrow r)$	$[(p \rightarrow q) \wedge (q \rightarrow r)]$	$(p \rightarrow r)$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
F	F	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	T	T	T	T	T	T	T
T	F	F	F	T	F	T	T
T	F	T	F	T	F	T	T
T	T	F	T	F	F	F	T
T	T	T	T	T	T	T	T

Problem 32

32 Show that $(p \wedge q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ are not logically equivalent.

$$\begin{array}{ll}
 (p \wedge q) \rightarrow r & (p \rightarrow r) \wedge (q \rightarrow r) \\
 \sim(p \wedge q) \vee r & (\sim p \vee r) \wedge (\sim q \vee r) \\
 \sim p \vee \sim q \vee r & \neq (\sim p \wedge \sim q) \vee r
 \end{array}$$

Problem 54

54 Show that $p|(q|r)$ and $(p|q)|r$ are not equivalent, so that the logical operator $|$ is not associative.

Section 1.4

Problem 14

14 Determine the truth value of each statement

- a)** $\exists x(x^3 = -1)$
- b)** $\exists x(x^4 < x^2)$
- c)** $\forall x((-x^2) = x^2)$
- d)** $\forall x(2x > x)$

- a)** True
- b)** True
- c)** True
- d)** False

Problem 36

Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all real numbers.

- a)** $\forall x(x^2 \neq x)$
- b)** $\forall x(x^2 \neq 2)$
- c)** $\forall x(|x| > 0)$

a) $\exists x(x^2 = x)$; Because when $x = 0$, $0^2 = 0$.

Therefore $x = 0$ is a counterexample

a) $\exists x(x^2 = 2)$; Because when $x = \sqrt{2}$, $(\sqrt{2})^2 = 2$.

Therefore $x = \sqrt{2}$ is a counterexample

c) $\exists x(|x| \leq 0)$; Because when $x = 0$, $|0| \leq 0$ is True.

Therefore $x = 0$ is a counterexample.

Section 1.5

Problem 4

Let $P(x, y)$ be the statement "Student x has taken class y ," where the domain for x consists of all students in your class and for y consists of all computer science courses at your school. Express each of these quantifications in English.

b) $\exists x \forall y P(x, y)$

c) $\forall x \exists y P(x, y)$

b) There is a student x who has taken every Computer Science class in my school.

c) For every student x this exists at least one Computer Science class in my school.

Problem 24

Translate each of these nested quantifications into an English statement that expresses a mathematical fact. The domain in each case consists of all real numbers.

a) $\exists x \forall y (x + y = y)$

b) $\forall x \forall y ((x \neq 0) \wedge (y \neq 0) \leftrightarrow (xy \neq 0))$

a) There exists at least one real number x such that the sum of its value and a real number y equal y .

b) For every pair x, y their product is non-zero if and only if x and y are also non-zero.

Problem 28

Determine the truth value of each of these statements if the domain of each variable consists of all real numbers.

a) $\forall x \exists y (x^2 = y)$

b) $\forall x \exists y (x = y^2)$

a) True

b) False

Problem 30

Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

b) $\sim \forall x \exists y P(x, y)$

c) $\sim \exists y (Q(y) \wedge \forall x \sim R(x, y))$

b) $\exists x \forall y \sim P(x, y)$

c) $\forall y (\sim Q(y) \vee \exists x R(x, y))$

Section 1.6

Problem 10

For each of these sets of premises, what relevant conclusion or conclusions can be drawn? Explain the rules of inference used to obtain each conclusion from the premises.

c) "All insects have six legs." "Dragonflies are insects." "Spiders do not have six legs." "Spiders eat dragonflies."

All insects have six legs.
Dragonflies are insects.
Spiders do not have six legs.
Spiders eat dragonflies.

Hypothetical Syllogism: Dragonflies have six legs.

Modus Tollens: Spiders are not insects.

Hypothetical Syllogism: Spiders eat all insects.

Conclusion: Spiders eat insects and dragonflies have six legs.

Problem 12

Show that the argument form with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$, and $\sim s$ and conclusion $q \rightarrow r$ is valid by first using Exercise 11 and then using rules of inference from Table 1.

r Modus Ponens

Section 1.7

Problem 8

8) Prove that if n is a perfect square, then $n + 2$ is not a perfect square.

Contradiction $p = n$ and $q = n + 2$

Assume $n = a^2$, and $n + 2 = b^2$

$$2 = b^2 - n$$

$$2 = b^2 - a^2 \text{ (Replacing } n \text{ by } a^2\text{)}$$

$$2 = (a + b)(a - b)$$

$$2 = (a + b) \text{ and } 1 = (a - b), \text{ which added together is } 2b = 3$$

$b = \frac{3}{2}$ which is not a perfect square.

c) Prove or disprove that the product of a nonzero rational number and an irrational number is irrational.

Contradiction

Let $p = \frac{a}{b}$ where $a \neq 0$, and $b \neq 0$, and k be an arbitrary irrational number.

$$\frac{a}{b} * k = \frac{ak}{b}$$

Section 1.8

Problem 12

Show that the product of two of the numbers $65^{1000} - 8^{2001} + 3^{177}$, $79^{1212} - 9^{2399} + 2^{2001}$, and $24^{4493} - 5^{8192} + 7^{1777}$ is nonnegative. Is your proof constructive or nonconstructive? [Hint: Do not try to evaluate these numbers!]

Let a , b , and c represent the three summations above. Since either a , b , or c can be a negative, positive, or zero value number, we have a few cases to consider.

If either of the two numbers is zero, then our product is zero; which is nonnegative.

If we chose 2 negative numbers, our product is a positive number.

For the case where we have a positive and negative number, we may end up with either a negative or positive number depending on our choice. Therefore, as long as we choose the right value, we can ensure that we always get a nonnegative value.

Non-Constructive

Problem 36

Prove that between every rational number and every irrational number there is an irrational number.

Let $r = \frac{a}{b}$ where $b \neq 0$, also let $\sqrt{2}r$ be an arbitrary irrational number.

Since we know that $r < \sqrt{2}r$, we can infer that $r < \frac{1}{2}\sqrt{2}r < \sqrt{2}r$

This proves that between every rational and irrational number, there is another irrational number.

Section 2.1

Problem 18

Find two sets A and B such that $A \in B$ and $A \subseteq B$

Let $A = \{0, 1, 2, 3\}$ and $B = \{0, 1, 2, 3, 4, 5, 6, 7\}$

Problem 20

What is the cardinality of each of these sets?

d) $\{\emptyset, \{\emptyset\}, \{\emptyset\}, \{\emptyset\}\}$

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Section 2.2

Problem 18

Let A , B , and C be sets. Show that

d) $(A - C) \cap (C - B) = \emptyset$

$(A - C) \cap (C - B) = \emptyset$

$(A \cap \overline{C}) \cap (C \cap \overline{B})$

Problem 32

Find the symmetric difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$

$\{2, 5\}$

Problem 48

Let $A_i = \{\dots, -2, -1, 0, 1, \dots, i\}$. Find

a) $\bigcup_{i=1}^n A_i$ **b)** $\bigcap_{i=1}^n A_i$

a) $\bigcup_{i=1}^n A_i = \{\dots, -2, -1, 0, 1, \dots, n\}$

b) $\bigcap_{i=1}^n A_i = \{\dots, -2, -1, 0, 1\}$